

EthiViz - Cultural Bias Analysis Platform

EthiViz is a comprehensive platform for analyzing cultural bias in both text and image data through multiple ethical traditions, including Western, Ubuntu, Confucian, and Islamic perspectives.

Features

- **Multi-tradition Ethical Analysis:** Analyze data through different cultural and ethical frameworks
- **Text & Image Analysis:** Process both textual and visual content for bias detection
- **Lightweight Image Processing:** Tiered feature extraction approach for computational efficiency
- **Interactive Visualizations:** Rich, interactive dashboards using D3.js and Plotly
- **Diversity Metrics:** Shannon entropy-based diversity calculations across multiple dimensions
- **Batch Processing:** Efficient processing of multiple images and text samples
- **Graceful Dependency Management:** Works with minimal dependencies but leverages advanced libraries when available
- **Streamlit UI:** Modern, sleek dark-themed user interface for intuitive analysis

Installation

Using the Setup Scripts (Recommended)

On Linux/macOS:

```
# Clone the repository
git clone https://github.com/yourusername/ethiviz.git
cd ethiviz

# Make the setup script executable and run it
chmod +x setup.sh
./setup.sh

# Activate the virtual environment
source activate-ethiviz.sh
```

```
# Run the application
streamlit run app.py
```

On Windows:

```
# Clone the repository
git clone https://github.com/yourusername/ethiviz.git
cd ethiviz
```

```
# Run the setup script
setup.bat
```

```
# Activate the virtual environment
activate-ethiviz.bat
```

```
# Run the application
streamlit run app.py
```

Manual Installation

```
# Clone the repository
git clone https://github.com/yourusername/ethiviz.git
cd ethiviz
```

```
# Create and activate virtual environment
python -m venv venv
source venv/bin/activate # On Windows: venv\Scripts\activate.bat
```

```
# Install dependencies
pip install -r requirements.txt
```

```
# Download spaCy model (required for text analysis)
python -m spacy download en_core_web_md
```

Usage

Streamlit Interface

The easiest way to use EthiViz is through its interactive Streamlit interface:

```
# Launch the Streamlit app
streamlit run app.py
```

This opens a web interface where you can:

- Upload text data (CSV, Excel) and images

- Select ethical traditions to include in the analysis
- Configure advanced settings like feature extraction level
- Run analyses and view interactive results
- Save results to a specified output directory

Command Line Interface

For batch processing or integration with other tools:

```
# Analyze text data
```

```
python run.py --text-data path/to/textdata.csv
```

```
# Analyze images
```

```
python run.py --image-dir path/to/image/directory
```

```
# Analyze both text and images
```

```
python run.py --text-data path/to/textdata.csv --image-dir path/
to/image/directory
```

```
# Specify ethical traditions to use
```

```
python run.py --text-data path/to/textdata.csv --traditions
western,ubuntu,confucian,islamic
```

```
# Set image analysis feature extraction level
```

```
python run.py --image-dir path/to/image/directory --feature-level
medium
```

```
# Specify output directory and dashboard port
```

```
python run.py --text-data path/to/textdata.csv --output-dir ./
my_results --port 8080
```

Python API

For programmatic use and custom applications:

```
from text_analyzer import TextAnalyzer
from image_analyzer import ImageAnalyzer
from visualization import create_dashboard
```

```
# Text analysis
```

```
text_analyzer = TextAnalyzer(traditions=['western', 'ubuntu', 'confucian',
'islamic'])
```

```
text_results = text_analyzer.analyze('path/to/data.csv')
```

```
# Image analysis
```

```
image_analyzer = ImageAnalyzer(feature_level='medium')
```

```
image_paths = ['path/to/image1.jpg', 'path/to/image2.jpg']
```

```
image_results = image_analyzer.analyze_batch(image_paths)

# Create dashboard
create_dashboard(text_results=text_results, image_results=image_results,
                 port=8050)
```

Ethical Traditions

EthiViz analyzes data through multiple ethical traditions:

1. **Western Ethics:** Based on principles like autonomy, beneficence, non-maleficence, and justice
2. **Ubuntu Ethics:** “I am because we are” - focuses on community, relationality, and shared humanity
3. **Confucian Ethics:** Emphasizes harmony, proper relationships, and virtues like benevolence and reciprocity
4. **Islamic Ethics:** Balances individual and community rights, emphasizing justice, fairness, and human dignity

Image Analysis

The image analysis functionality includes:

- **Skin Tone Analysis:** Fitzpatrick scale approximation to analyze representation diversity
- **Feature Extraction:** Tiered approach with color histograms, HOG features, and deep learning
- **Diversity Calculations:** Shannon entropy-based metrics for skin tone, gender, age distribution
- **Cultural Element Detection:** Identification of cultural signifiers and their distribution

Visualizations

The platform provides rich interactive visualizations:

- **Metric Cards:** At-a-glance summary of key metrics
- **Comparative Charts:** Compare scores across ethical traditions
- **Demographic Analysis:** Visualize demographic distributions in images
- **Radar Charts:** Multi-dimensional ethical analysis
- **Image Gallery:** View analyzed images with their metrics
- **Combined Analysis:** Compare text and image results

Requirements

- Python 3.8+
- NumPy, Pandas, SciPy
- Streamlit, Plotly (for visualization)
- Pillow (required for basic image processing)

- OpenCV (optional - enables medium-level feature extraction)
- TensorFlow (optional - enables advanced feature extraction)
- SpaCy (with at least en_core_web_md model)